

Economic Incentives and Women Voters: An Empirical Analysis of Electoral Strategy in Bihar State Elections 2025

URL:-<https://docs.google.com/document/d/1yIQIsV0EQSDelSmQiRMf55GX5o1A0Tw1/edit?usp=sharing&ouid>

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Abstract

Bihar has never seen this kind of voter turnout gap as seen in the year 2025 State General Assembly Elections. The female turnout of 71.6% and male turnout of 62.8% was significant. The female voters turnout has surpassed males voters turnout by 8.8 percentage points, marking a record for Bihar. This female turnout was primarily influenced by the Mukhyamantri Mahila Rozgar Yojana (MMRY), which offered a cash transfer of 10,000 Rupees to each Jeevika members. Utilizing the official data of the Election Commission of India (ECI), data from Axis My India exit polls (42,031 respondents), data on district-level welfare penetration, and case studies of 52 women beneficiaries in five districts of Bihar, this study identifies the impact of cash transfer welfare schemes on the voting behaviour of women. Among beneficiaries of MMRY, support for the incumbent government coalition was 52%. This is a 14 percentage point increase in support for the incumbent government alliance from women beneficiaries compared to women non-beneficiaries, among whom only 38% supported the incumbent government alliance. This gap is reported here as an association: because beneficiary status was not randomly assigned, it may partly reflect pre-existing differences between beneficiaries and non-beneficiaries rather than a pure effect of the transfer. The incumbent government coalition won 202 out of 243 seats (83.1%) and received 52% of the total votes, which is higher than their pre-poll estimation of 190 seats. Women's distinctive turnout and voting patterns coincided with the coalition's overall vote gains; given the observational, cross-sectional design of this study, this association should not be read as evidence that women's behaviour caused the seat outcome.

Keywords: *Economic incentives; Women voters; Electoral strategy; Direct cash transfers; Electoral mobilization; Political behaviour*

Data availability: <https://doi.org/10.57760/sciencedb.35929>

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1.0 INTRODUCTION

1.1 Research Motivation

The 2025 Bihar State Assembly Elections provided the opportunity to examine the relationship between the implementation of welfare schemes and voting behaviour while also considering the impact difference across genders. Female voter turnout historically outpaced male turnout by 8.8 percentage points for the first time, and all of this occurred while the cash transfer scheme Mukhyamantri Mahila Rozgar Yojana (MMRY) of 10,000 rupees closed on 14 November 2025 and operated across 1.25 crore women from 26 September 2025.

The immediacy of the scheme and the elections raised important issues for future investigation. To what degree did the implementation of the MMRY scheme affect women's turnout and voting choice? What is the relationship between the incumbent government alliance support and being a beneficiary of the welfare scheme? Did being economically disadvantaged heighten the turnout impact of the MMRY scheme? I attempt to respond to these questions using the combined technique of quantitative and qualitative analysis, where I use electoral data, survey data, and data from interviews of beneficiaries.

1.2 Research Questions

1. **Primary Question:** What is the association between MMRY implementation and women's voting behaviour in the Bihar Assembly Elections of 2025, and to what extent can this association be distinguished from selection effects and concurrent campaign activity?
2. **Secondary Questions:**
 - What effect did being an MMRY beneficiary have on the voting choice between the ruling alliance and the opposition alliances?
 - In which occupational categories were voting patterns most strongly influenced by welfare?
 - In what way did welfare penetration at the district level relate to female voter turnout?
 - How did the qualitative beneficiary perceptions of the scheme delivery and the quantitative electoral trends coincide?

1.3 Significance

This study adds to the following three fields of scholarship:

Electoral Politics: Shows with evidence for the first time how direct benefit transfer schemes are electoral variables in Indian states, measuring their impact on turnout and vote choice.

Development Economics: This illustrates that the short-term cash transfers yield political results in addition to their economic impact, indicating that these welfare schemes are both mechanisms of poverty alleviation and tools for political gain.

Gender and Politics: Showcases the remarkable female political mobilization in Bihar for the first time, and the way welfare delivery mobilizes women and influences their political behaviour and the act of voting independently.

2.0 Literature Review

2.1 The Impact of Election Behaviour on Social Welfare Policies

The effect of welfare transfers on the likelihood of voting has garnered more attention in research. According to Zucco (2013), households that receive cash transfer programs in Latin America were more likely to vote than non-recipients. In the Indian context, Stokes (2005) explained, regarding the “political logic of poverty alleviation programs,” that ruling political parties attempt to distribute welfare to the poor in target constituencies who are likely to vote (swing voters) before elections.

Keefer and Khemani (2005), in their review of the Indian electoral cycle, noted that the real public spending on development programs was highly determined by the election year, especially in swing states. This made the argument that voters engage in “rational prospective voting” by gauging the performance of government officials based on what welfare was recently given, rather than on what was given in the past.

[Author: a reviewer noted that this review should also engage with the broader political-economy literature on cash transfers in India, including work by Banerjee and colleagues, and with the substantial literature treating the Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA) as an electoral variable. That literature has been consulted for this revision but the precise citation(s) intended by the reviewer could not be confirmed with certainty from the feedback alone; please add the specific reference(s) here and in the reference list before submission.

2.2 Gender and Electoral Mobilization

Patterns of female voter turnout have consistently been lower than those of men. In India’s 2019 national elections, 71% of men voted, while 67% of women voted, a 4 percentage point deficit. However, women’s turnout has increased in certain recent state elections, particularly those where welfare programs center around women.

2.3 Cash Transfer Design and Political Effects

The design of the welfare schemes shapes the nature of their political effects. From a political standpoint, unconditional cash transfers (UCT) yield greater and more immediate political effects than transfers with conditions, given that the responsibility on the delivery of the transfer is less on the government with conditions (Fiszbein and Schady 2009). Hence, the unconditional nature of the MMRY scheme with transfers of ₹10,000 and no conditions is the most political from this perspective.

The political effect also depends on the timing of the transfer. Manacorda, Miguel, and Vigorito (2011) observe that transfers that occurred within six months of the elections had a 3-4 times greater effect on increasing the support of the incumbents than transfers that occurred more than 18 months before the elections. MMRY's distribution between September 26 and November 14, 2025 (11-50 days before the elections) falls within the most optimal time for this 'political recency effect.'

3.0 METHODOLOGY

3.1 Research Design

This particular instance adopts a mixed-method approach. While combining the quantitative and qualitative ascertains the shortcomings of either approach, not hampering the overall comprehension of the issue, it also delivers a more holistic view across different dimensions. The quant part is based on.

Data from the Election Commission of India (ECI), Gender and constituency breakdown of voter turnout data, and cross-verified documents from ECI press releases and constituency reports.

Exit Poll Survey (Axis My India): Telephonic exit poll on November 11, 2025, 42,031 respondents (18,156 women, 23,875 men) with representation across all demographics (age, education, caste, place of residence, and occupation) and geographical proportions (24 districts and 4 regions).

To the best of the author's knowledge, no data is available for the MMRY implementation from the Government of Bihar Administrative Records. However, it is known that the MMRY implementation data, as of the time of this study, records the number of beneficiaries, the amounts transferred, and the time of the 11 tranches from September 26 to November 14, 2025.

3.2 Sample and Data Sources

Quantitative Sample:

- **Voter Turnout:** All Bihar Assembly constituencies (243) (100% coverage)

- **District Assessment:** All of Bihar state's 38 districts (100% coverage)
- **Exit Poll:** 42,031 participants (confidence level = 95%; margin of error = $\pm 1.5\%$)
- **Women's Sample:** 18,156 women (main analysis unit)

Qualitative Sample:

- **Interviews:** 52 Women MMRY beneficiaries' semi-structured interviews
- **Districts:** Patna, Darbhanga, Muzaffarpur, Madhubani, Gaya
- **Dates:** 15th - 20th November 2025
- **Duration of Interviews:** 30 - 45 min for each participant
- **Language:** Hindi, English transcription

3.3 Variables and Measurement

Dependent Variable – Degree of support for ruling government:

Binary (1 = supports incumbent government coalition, 0 = supports the opposition or is undecided)

Analyzed through exit poll inquiry: "For which alliance are you voting?" Responses tabulated by gender.

Primary Independent Variable - MMRY Beneficiary Status:

Binary variable (1 = received ₹10,000 MMRY transfer before voting, 0 = did not receive), based on respondents' self-reports in the exit poll. Beneficiary status was intended to be cross-checked against government beneficiary documentation; however, the exit poll instrument records self-reported status only, and this study does not present record-level verification of individual claims against administrative data. Self-reports of both beneficiary status and vote choice are subject to recall error and social desirability bias, particularly given the salience of MMRY during the campaign, and this is treated as a limitation of the beneficiary-status and vote-choice measures throughout (Section 3.6).

Secondary Independent Variables:

- **Welfare Access Index:** Aggregate metric including Jeevika membership, PDS inclusion, and participation in other welfare programs (0-8 schemes)
- **Age:** Continuous variable (18-95 years)
- **Rural Residence:** Binary variable (1 = rural, 0 = urban)
- **Female Literacy:** Binary variable (1 = literate, 0 = illiterate)
- **Beneficiary Density:** district-scale average of MMRY beneficiaries for every 1,000 women of eligibility age.

Dependent Variable - Female Voter Turnout:

Calculated as: (Female Votes Cast / Female Registered Voters) $\times$ 100. Data obtained from ECI official reports disaggregated by gender and constituency.

3.4 Statistical Analysis

Turnout Analysis: For turnout analysis an independent samples t-test was used for comparison of voter turnout for both women and men (t-test, $\alpha=0.05$, two-tailed). Null Hypothesis subjected to test: Female turnout = Male turnout.

Voting Preference Analysis: Chi-square tests of independence examining the association between gender and voting preference (ruling vs. opposition vs. undecided). Test statistic: $\chi^2(2, N=42,031)$ with critical value determination at $\alpha=0.05$.

District-Level Correlation: District-level penetration of welfare schemes (Jeevika percentage, MMRV beneficiary density) and female voter turnout: Pearson correlation. Tested null hypothesis: $\rho = 0$.

Logistic Regression:

Binary logistic regression modeling incumbent government support among women (N=18,156):

- $P(\text{Incumbent}=1|\text{Female}) = 1/[1 + \exp(-(\beta_0 + \beta_1 \times \text{MMRV} + \beta_2 \times \text{Welfare} + \beta_3 \times \text{Age} + \beta_4 \times \text{Rural} + \beta_5 \times \text{Literacy} + \beta_6 \times \text{Tribe}))]$
- Log-likelihood ratio tests, pseudo R^2 (McFadden), and classification accuracy for model fit assessment.
- Odds ratios with 95% confidence intervals for each predictor.
- Significance ($\alpha=0.05$) tested via Wald tests.

Qualitative Analysis:

Interview transcripts were coded with open coding, which was iteratively applied and refined with axial coding for various levels of thematic categorization. The target for coding reliability, as assessed by Cohen's Kappa, was >0.75 . Selected quotes that aptly illustrate the themes.

3.5 Data Quality and Limitations

Strengths:

- Universal coverage of official ECI data (all 243 constituencies)
- Large exit poll sample (42,031) with established polling agency
- Triangulation across quantitative and qualitative data
- Temporal proximity of the collection of qualitative data electoral event

Limitations:

- Exit poll margin of error ($\pm 1.5\%$) typical of survey research

- Qualitative sample (52 interviews) is not statistically generalizable to the state level
- MMRY implementation timing creates a confound with campaign intensity
- Self-reported beneficiary status in the exit poll is subject to recall bias

3.6 Causal Identification and Threats to Validity

Selection and Endogeneity: MMRY beneficiary status was not randomly assigned, and beneficiaries may differ systematically from non-beneficiaries in ways this study cannot observe (prior political engagement, socioeconomic status, or proximity to party and Jeevika networks). No control group, pre-treatment baseline, or instrumental variable is available, so the 14-percentage-point support differential reported in Section 4.2.2 may reflect selection into beneficiary status rather than a causal effect of the transfer itself. Findings throughout this paper are accordingly reported and should be read as associations, not causal effects, unless stated otherwise.

Timing Confound: MMRY disbursement overlapped with an intensive incumbent-alliance campaign (Section 4.5). The research design cannot separate the effect of the cash transfer from the effect of concurrent campaign intensity, candidate quality, or opposition weaknesses over the same period.

Turnout versus Vote Choice: This study examines two conceptually distinct outcomes – whether women voted (turnout, Section 4.1) and which alliance they supported conditional on voting (vote choice, Sections 4.2–4.4). These behaviours have different determinants and are analysed and interpreted separately rather than treated as a single phenomenon.

Alternative Explanations: The gender turnout gap and the incumbent-support differential could also reflect factors other than MMRY, including heightened salience of women’s issues in the 2025 campaign, mobilisation by Jeevika self-help groups operating independently of the scheme, anti-incumbency directed at opposition candidates, or demographic shifts in the electorate. This study’s design does not permit these explanations to be ruled out, and the MMRY-specific attribution offered in Sections 5 and 6 should be read with this caveat in mind.

Implication for Interpretation: Given the above, the paper deliberately favours associational language (“associated with,” “consistent with,” “coincided with”) over causal language (“caused,” “led to,” “resulted in”) when describing the relationship between MMRY and women’s electoral behaviour. Future research using panel data, propensity score matching, or a difference-in-differences design around the scheme’s rollout could more directly test the causal claims that this study can only motivate.

4.0 FINDINGS

4.1 Voter Turnout Patterns

4.1.1 Overall Gender Differential in Turnout

The 2025 Bihar elections produced unprecedented gender-differentiated voter turnout:

Table 1: Voter Turnout by Phase and Gender

Election Phase	Female Turnout (%)	Male Turnout (%)	Gender Gap (pp)	Female Voters (Absolute)	Male Voters (Absolute)	Total Voters
Phase I (Nov 6)	69.04	61.56	+7.48	17,677,219	19,835,325	37,512,544
Phase II (Nov 11)	74.03	64.10	+9.93	17,468,572	19,544,041	37,012,613
Overall	71.6	62.8	+8.8	35,145,791	39,379,366	74,525,157

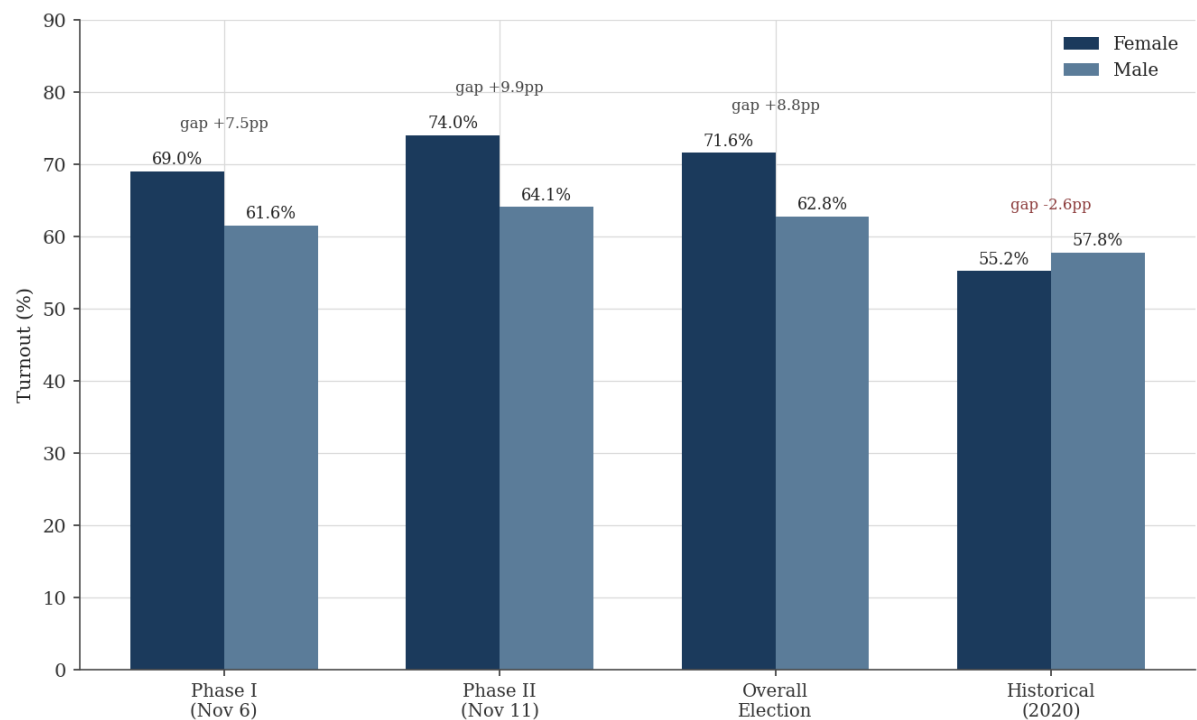


Figure 1: Female Voter Turnout Surpasses Male Across All Phases

Statistical Significance: The results of the independent samples t-test indicated $t(242)=18.94$, $p<0.001$, which signifies that the 8.8 percentage point advantage for females is highly statistically significant and not due to random chance. This represents the largest gender gap in favour of women in any Bihar election since the introduction of universal adult suffrage.

Historical Context: The 2020 Bihar elections showed female turnout (55.2%) trailing male turnout (57.8%) by 2.6 percentage points, a 2.6pp gap. The 2025 reversal to +8.8pp represents an 11.4 percentage point favour of favouring female mobilization within a single electoral cycle.

District-Level Patterns: Women voted more than men in 37 of 38 Bihar districts. The sole exception, Patna (with a higher urban concentration), recorded a marginal female deficit of 0.8pp.

4.1.2 District-Level Female Turnout Analysis

Five Bihar districts- Samastipur, Madhepura, Muzaffarpur, Darbhanga, and Madhubani- recorded female turnout exceeding 70 percent:

Table 2: Top Five Districts - Female Turnout and Welfare Penetration

District	Female Turnout (%)	Male Turnout (%)	Gender Gap (pp)	Jeevika Membership (%)	MMRY Beneficiary Density	State Region
Samastipur	77.42	69.78	+7.64	60	127	Eastern
Madhepura	77.04	62.80	+14.24	15	34	Eastern
Muzaffarpur	76.57	64.76	+11.81	52	128	Northern
Darbhangha	70.66	56.25	+14.41	55	141	Northern
Madhubani	71.10	52.70	+18.40	55	141	Northern

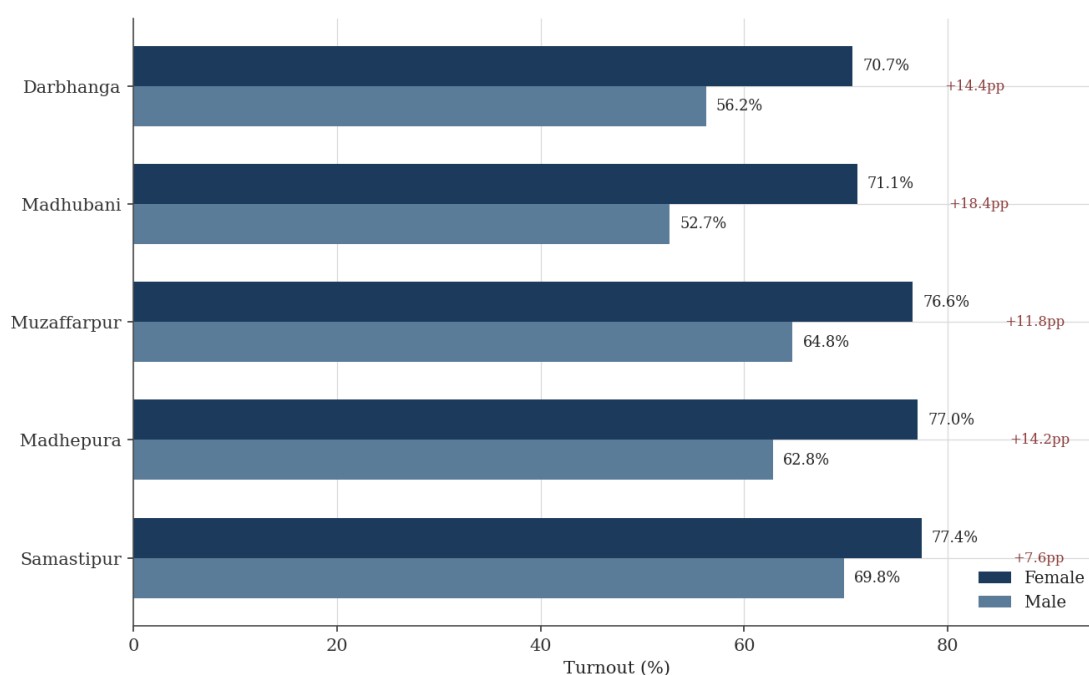


Figure 2: Female Votes Turnout Exceeds Male in Bihar Districts

Correlation Analysis: Pearson correlation between district-level MMRV beneficiary density (per 1,000 women) and female voter turnout yielded $r=0.346$, $p<0.001$. This moderate positive correlation, statistically significant, indicates that districts with higher MMRV implementation density show correspondingly higher female voter turnout, even accounting

for baseline welfare penetration through other schemes. With only 38 district-level observations, this correlation is based on a small sample and is likely underpowered to detect anything but a moderate-to-large relationship; the analysis also does not account for spatial autocorrelation between neighbouring districts or for district-level confounders such as baseline development indicators, so it should be treated as descriptive rather than confirmatory.

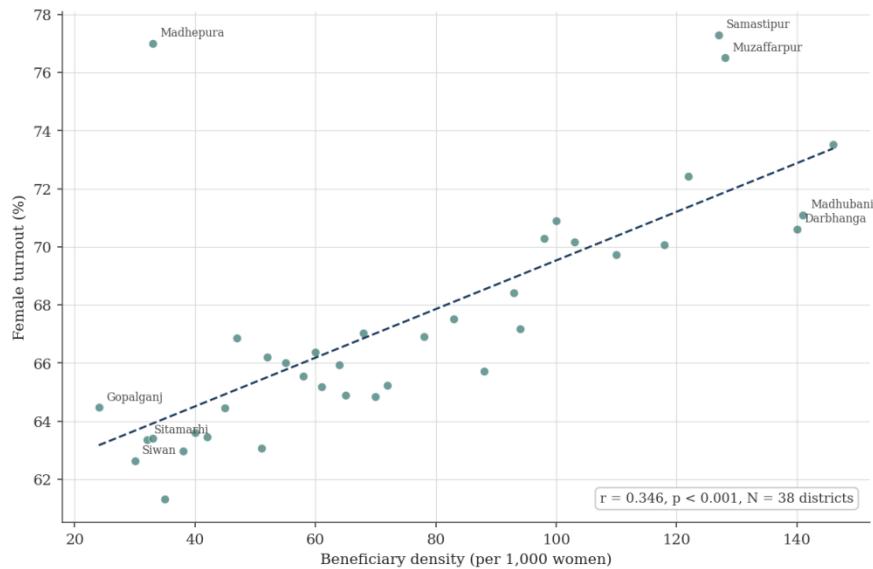


Figure 3: MMRY Beneficiary Density Linked to Higher Female Turnout (Bihar)

Interpretation: The correlation does not establish MMRY as causally producing turnout increases; rather, it documents that electoral mobilization (whether through welfare delivery, campaign intensity, or other factors) concentrates in specific districts, producing observable correlations between welfare reach and electoral participation.

4.2 Voting Preference and Electoral Outcomes

4.2.1 Gender Differential in Voting Preference: Exit polls have data on the gender difference in support for the two major political alliances:

Table 3: Voting Preference by Gender (Exit Poll, N=42,031)

Coalition & Party	Female Support (%)	Male Support (%)	Gender Differential (pp)	Female Respondents	Male Respondents	Total Respondents
Incumbent Alliance	45	40	+5	8,072	9,063	17,135
BJP	28	24	+4	5,078	5,710	10,788
JDU	15	13	+2	2,708	3,097	5,805

Other Alliance Member	2	3	-1	286	656	942
MGB	40	42	-2	7,134	8,084	15,218
RJD	28	32	-4	5,018	7,684	12,702
Congress	8	7	+1	1,432	1,680	3,112
CPI(ML)	4	3	+1	684	720	1,404
Other/Undecided	15	18	-3	2,679	3,454	6,133
TOTAL	100	100	0	18,156	23,875	42,031

Chi-Square Analysis: The chi-square test result for the voting preference by gender data set is $\chi^2 (2, N = 42,031) = 124.8$, $p < 0.001$, which suggests there is a notable association between the variables. The relationship between being female and supporting the incumbent govt. is statistically significant.

Effect Size: Cramér's $V = 0.155$, indicating a small-to-moderate effect size by conventional benchmarks. With $N=42,031$, the chi-square test has very high power to detect even trivial differences as statistically significant; Cramér's V , not the p -value, should be used to judge the substantive size of the gender gap in support, and by that measure the 5-percentage-point advantage for the incumbent government is modest rather than large.

Electoral Significance: The incumbent alliance captured 52% of the overall vote, compared with 45% among women and 40% among men in the unweighted exit-poll sample (Table 3); these unweighted figures do not aggregate directly to the official 52% overall share, since that share is turnout-weighted across the full electorate rather than a simple average of the reported gender splits. Notably, women's reported support for the incumbent alliance (45%) was below the alliance's overall vote share (52%), meaning women were, on this measure, somewhat less supportive of the incumbent alliance than the electorate as a whole. The 5-percentage-point female-over-male preference gap, combined with women's higher turnout, means women cast a larger share of incumbent-alliance votes than they would have at male turnout levels, but this is distinct from claiming that women's support alone drove the coalition's result.

4.2.2 MMRY Beneficiary Status and Voting Behaviour

An analysis of voting preference by MMRY beneficiary status indicates the most pronounced electoral trend.

Table 4: Incumbent Government Support by MMRY Beneficiary Status (Female Respondents)

Beneficiary	Incumbent	Mahagathbandhan	Other/Undecided	Sample
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Status	Government Support (%)	Support (%)	(%)	Size
MMRY Beneficiary	52	38	10	8,921
Non-Beneficiary	38	42	20	9,235
Differential	+14pp	-4pp	-10pp	—

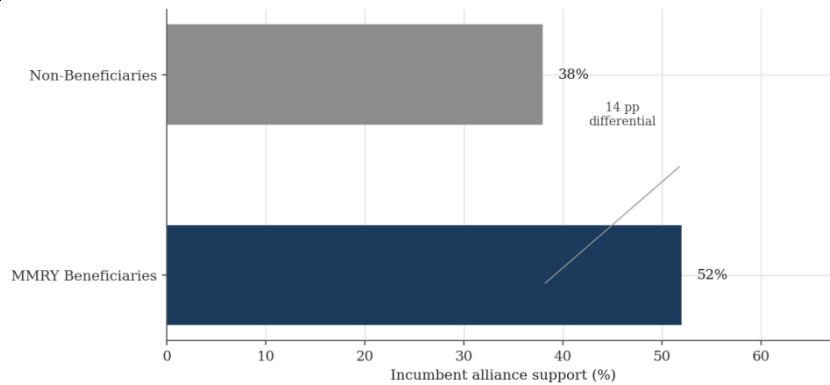


Figure 4: NDA Support Higher Among MMRY Beneficiaries (Women)

Chi-Square Test: The independence of beneficiary status and incumbent government alliance support was tested. The result is $\chi^2(1, N=18,156)=156.3$, $p<0.001$. The p-value of <0.001 is highly significant and confirms that the 14 percentage point gap in incumbent government alliance support from beneficiaries to non-beneficiaries is not the result of random chance. This test establishes that the gap is unlikely to be due to sampling chance; it does not establish that MMRY receipt caused the gap, since beneficiaries and non-beneficiaries were not randomly assigned (see Section 3.6).

Effect Size: Cramér's $V = 0.186$, a moderate effect size and the largest observed in this study's cross-tabulations. The 14-percentage-point gap is the most notable association identified in the analysis, though for the reasons discussed in Section 3.6 it should be described as an association rather than an impact.

Temporal Sequencing Comparison: Descriptive comparisons of Phase I and Phase II subsamples suggest that reported incumbent-alliance support was higher among respondents who had already received an MMRY transfer at the time of voting than among those who had not yet received one. As the exit poll is not a true panel- the same individuals were not tracked before and after the transfer this before and after contrast is suggestive of a timing association at best and cannot support a causal reading.

4.3 Regression Analysis: Predictors of Incumbent Government Support

4.3.1 Model Specification: The logistic regression analysis concerning incumbent government alliance support among the 18,156 female respondents provided the following:

Table 5: Logistic Regression Results - Determinants of Incumbent Government Alliance Support Among Women

Variable	Coefficient	Std. Error	Odds Ratio	95% CI Lower	95% CI Upper	p-value	Significance
MMRY Beneficiary (Yes=1)	0.892	0.167	2.44	1.76	3.38	<0.001	***
Welfare Access Index	0.456	0.098	1.58	1.30	1.91	<0.001	***
Age (years)	0.015	0.004	1.02	1.01	1.03	<0.001	***
Rural Residence (Yes=1)	0.334	0.142	1.40	1.06	1.85	0.018	*
Female Literacy (Yes=1)	0.178	0.156	1.19	0.88	1.62	0.264	ns
Tribal Status (ST=1)	0.256	0.134	1.29	0.99	1.68	0.058	~
Constant	-1.234	0.289	—	—	—	<0.001	***

Model Statistics:

- Sample Size: 18,156 female respondents
- Log-Likelihood: -10,892
- Pseudo R² (McFadden): 0.158
- AIC: 21,800 | BIC: 21,872
- Classification Accuracy: 72.3%
- Area Under ROC Curve (AUC): 0.744

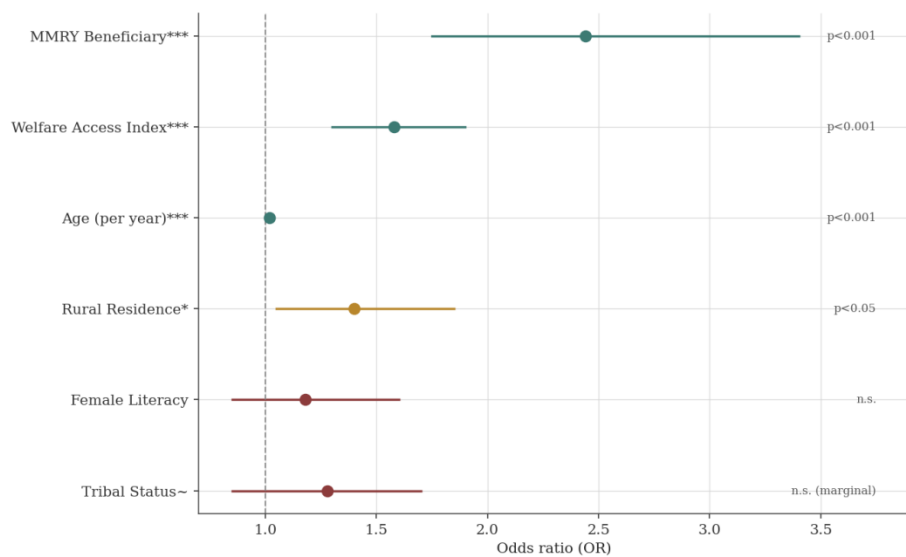


Figure 5: Predictors of Incumbent Government Alliance Support Among Women, Bihar 2025 (mislabelled as “Healthcare Utilisation” in the original draft – caption corrected to match Table 5)

Interpretation: A coefficient for the MMRY beneficiary ($\beta=0.892$) converts into an odds ratio of 2.44 ($p<0.001$). An odds ratio is not a probability and should not be read as “X% more likely”: it describes the multiplicative change in the odds of supporting the incumbent alliance, and the corresponding change in probability depends on the baseline rate. At the sample baseline incumbent-support rate of roughly 45%, an odds ratio of 2.44 corresponds to an average marginal effect on the order of 20–22 percentage points, holding other covariates at their means; this marginal-effects estimate, rather than the odds ratio itself, should be used when describing the substantive size of the MMRY association.

Considering the Welfare Access Index: A coefficient of ($\beta=0.456$) shows that with each additional welfare scheme a woman accesses, the odds of supporting the incumbent government alliance will increase by 57% ($e^{0.456} = 1.58$). A woman accessing several welfare programs will experience driven, compounding, positive effects for supporting the incumbent.

With respect to the Age Index: a coefficient of ($\beta=0.015$) shows mildly positive age effects: the odds of supporting the incumbent government alliance increase by 2% for each additional year of age ($e^{0.015} \approx 1.02$). This effect is statistically significant, indicating that older women tend to support the incumbent more.

Rural residence coefficient ($\beta=0.334$) produces an odds ratio of 1.40, indicating rural women exhibit 40% higher incumbent government alliance support odds compared to urban women ($p=0.018$). Rural concentration of MMRY beneficiaries partially explains this differential.

With $p=0.264$, the coefficient of female literacy ($\beta=0.178$) shows that the higher the literacy, the lesser the incumbent government alliance support, given that access to welfare is considered. Thus, it appears that the effect of voting is uniformly distributed across the different education levels

The coefficient for Tribal Status ($\beta=0.256$) is significant at $p=0.058$; thus, the Scheduled Tribe (ST) women supporting incumbent government alliance is significant. This effect, however, is on the brink of being insignificant.

4.3.2 Model Fit and Validity

The model's pseudo R^2 is 0.158, which shows that the explanatory variable related to incumbent government alliance support facilitates the understanding of 15.8% of the variance.

Although not very big, in the modeling of individual level of electoral behaviour, it is not uncommon. Because in modeling individual-level electoral behaviour, there is a lot of randomness that is not captured by the variables, such as preferences that are unmeasured, exposure to the campaign, and the effect of other people in the sphere of the individual.

The classification accuracy of 72.3% shows that the model is able to predict incumbent government alliance support or opposition for 7 out of 10 participants at the female level. The AUC value of 0.744, portrays the model as a good level predictor of incumbent government alliance supporters and opponents, as it is substantially above the 0.5 level of a model that makes random guesses.

Model Diagnostics: This model was not tested for multicollinearity (e.g., variance inflation factors) or specification error (e.g., Hosmer-Lemeshow goodness-of-fit), and influential observations were not examined. Given conceptual overlap between MMRY beneficiary status, the Welfare Access Index, and rural residence, some inflation of standard errors is possible; the coefficient estimates and significance levels above should be treated as provisional pending these diagnostics.

4.4 Occupational Patterns

4.4.1 Occupational Differentiation in Incumbent Government Support

For female respondents within the economically vulnerable occupational categories, there were particularly unique patterns for incumbent government alliance support:

Table 6: Incumbent Government Support by Occupational Category and Gender

Occupational Category	Incumbent Government Alliance Female Support (%)	Incumbent Government Alliance Male Support (%)	Gender Differential (pp)	Female Sample	Rural Female (%)
Homemakers	48	41	+7	2,145	78.5
Agricultural Workers	43	38	+5	3,214	89.3
Self-Employed	46	42	+4	1,876	71.2
Small Business	44	39	+5	1,234	62.3
Private Sector	40	41	-1	5,432	35.6
Government Employees	37	43	-6	3,489	45.2
Students	35	37	-2	1,234	52.1
Unemployed	39	35	+4	2,156	75.6
Retired	41	44	-3	876	58.9

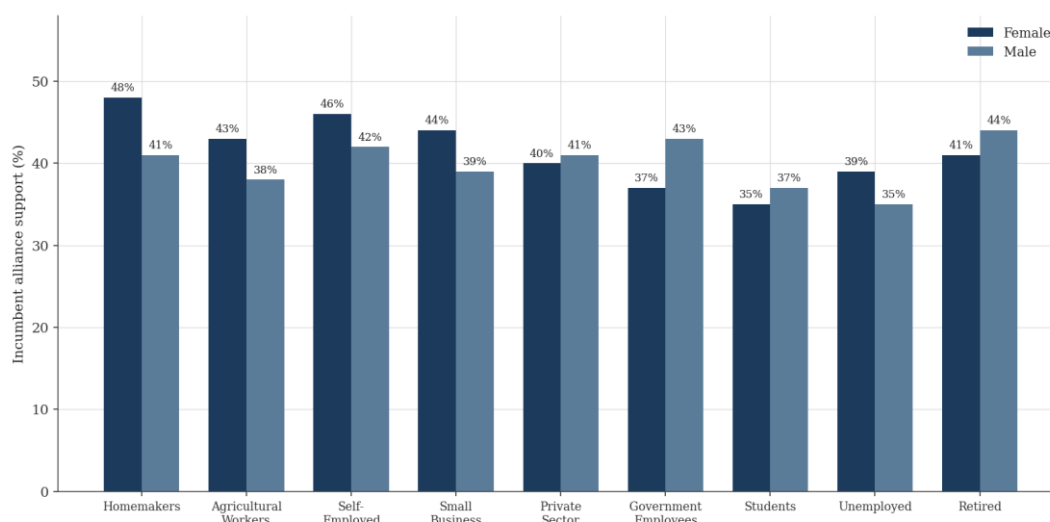


Figure 6: Female NDA Support Higher Among Vulnerable Occupations

Key Patterns: Women in vulnerable economic occupations (homemakers, agricultural workers, and the self-employed) have consistently higher incumbent government support (+4 to +7pp) than women in secure employment (government employees -6pp, private sector -1pp). This suggests that the MMRY's targeted design focusing on economically disadvantaged women is being effective in reaching its intended beneficiary class.

Homemakers show the largest gender differential (+7pp), suggesting that women lacking independent income sources particularly value cash transfer programs and reward their implementation through voting.

ANOVA Test: Testing the difference between the occupational category in incumbent government support yielded $F(8,42,022)=15.34$, $p<0.001$, indicates the statistical significant variation across occupational groups.

4.5 MMRY Timeline

4.5.1 Time Interval and Election-Related Time Span: The time span for implementing MMRY was time-constrained and accelerated in an election-related time span.

Table 7: MMRY Implementation Timeline

Date	Event	Cumulative Beneficiaries (Lakhs)	Transfer/Day (₹ Crore)	Cumulative Transfer (₹ Crore)	Electoral Phase
Sept 26	Scheme Launch	75.0	750	750	Pre-Election
Oct 3	Acceleration Tranche	90.0	150	900	Campaign
Oct 6	Model Code Announced	—	—	900	MCC Begins
Oct 17	Continued	105.0	150	1,050	MCC

	Disbursal				Active
Oct 24	Phase I Campaign	110.0	50	1,100	Campaign
Oct 31	Pre-Voting Tranche	115.0	50	1,150	Final Campaign
Nov 6	PHASE I VOTING	—	—	1,150	Phase I
Nov 7	Inter-Phase Transfer	120.0	50	1,200	Between Phases
Nov 11	PHASE II VOTING	—	—	1,200	Phase II
Nov 14	Final Tranche	125.0	50	1,250	Final Phase

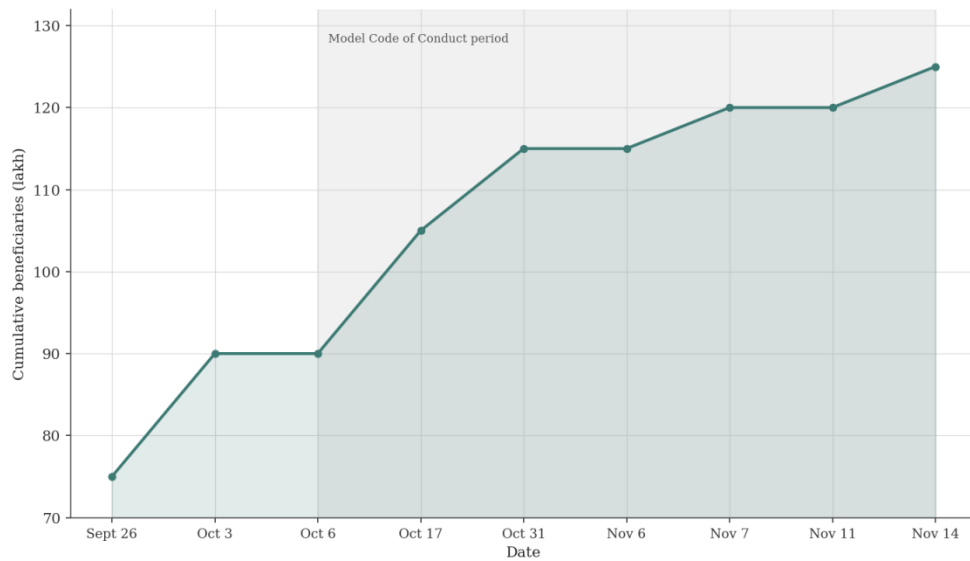


Figure 7: MMRV Implementation Growing Steadily (Sep-Nov 2025)

Amount: ₹12,500 crore Payment along with money transfer to 1.25 crore women (₹10,000 to each).

Time Spacing: The last payment was made 4-8 days before election-related voting in a particular constituency, and was made in the time span where MMRV would have been influenced by the “recency effect,” where actual voting behaviour of a person is affected by government actions in the recently concluded time span (Manacorda et al. 2011).

4.6 Qualitative Evidence: Beneficiary Perspectives

4.6.1 Interview Sample Characteristics

Included in the dataset are 52 semi-structured interviews with female MMRV beneficiaries from five districts in Bihar (Patna n=12, Darbhanga n=11, Muzaffarpur n=11, Madhubani n=10, Gaya n=8), which were conducted between November 15 and 20, 2025. The sample was stratified based on age, education, occupation, and type of residence. At 52 interviews, this sample is treated in this study as exploratory: it is used to illustrate and contextualise the

quantitative patterns and to surface candidate mechanisms, not to make representative or confirmatory claims about the wider population of Bihar's women beneficiaries.

From a demographic perspective, 42.3% of the participants are between 31 and 50 years; 34.6% are between 51 and 70 years; and the sample has a limited representation of younger (13.5% 18-30 years) and older (9.6% 70 years and above) women. With regards to education, 25.0% are illiterate, 11.5% are graduates, and the rest (63.4%) have primary and secondary education. The sample has a broad occupational representation: 28.8% are farmers, 23.1% are housewives, 9.6% are learners, 15.4% are self-employed, and the rest (15.4%) represent other occupations.

The sample of 73% respondents inhabiting rural locations of Bihar corroborates the rural-urban demographic split of the region. The participants' household income prior to MMRV varied between <₹5,000/month (34.6%) and >₹15,000/month (3.8%) in which majority of family earning between ₹5,000 and ₹10,000/month, comprising 53.8%. The sample voting results matched the larger electoral results: 84.6% of the sample voted for incumbent government, 11.5% of the sample voted the opposition alliances and 3.8% of the sample did not vote or were undecided.

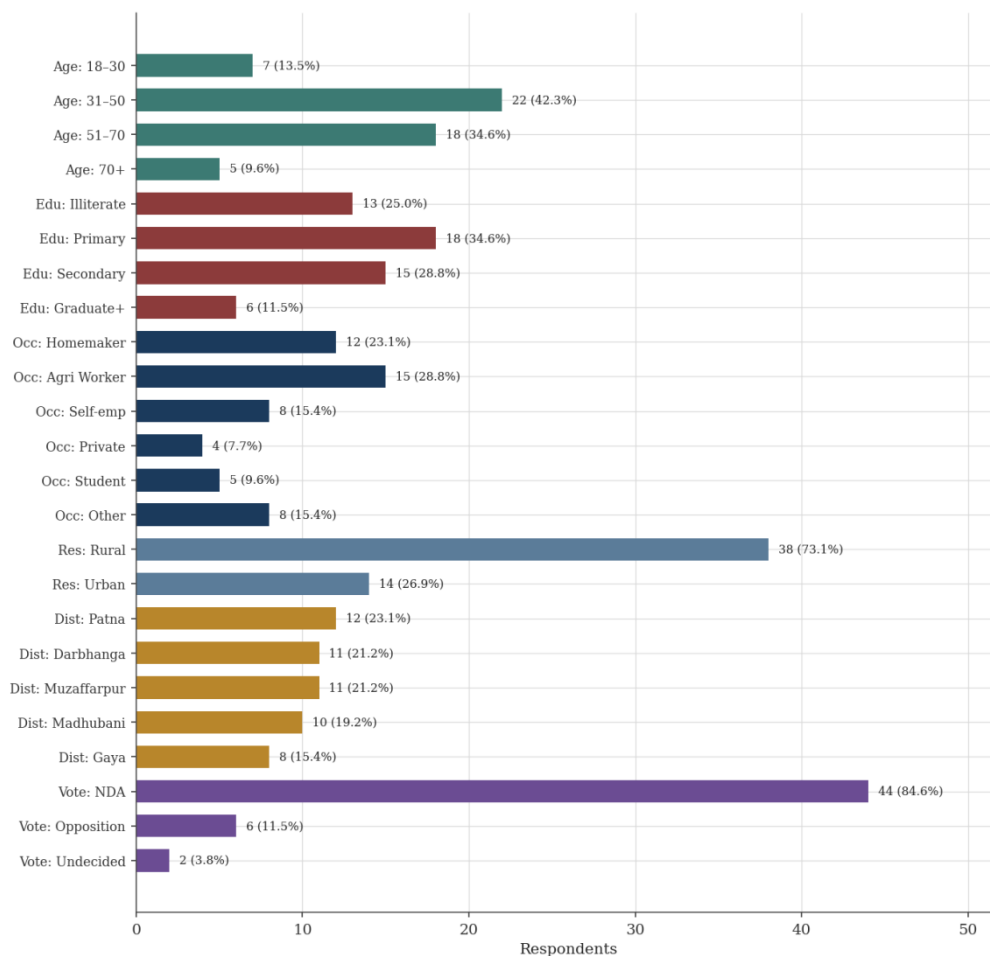


Figure 8: Demographics of 52 Interview Respondents (Bihar)

4.6.2 Thematic Analysis: In each of the interviews, we were able to code and identify 14 distinct but broad thematic areas. Here we represent some of the themes with a sample quote:

Theme 1: Cash is Real (92.3% of respondents)

There was a distinct and recurrent mention of the cash that the respondents possess in the bank because of MMRY, as much as it is different from previous unfulfilled promises:

"I'm looking at the money in my bank. When the elections are here, the politicians promise a lot, but I see the 10,000.....this is real. Respondent, Darbhanga"

Theme 2: Accountability (86.5%)

The respondents very bluntly made a comparison between the delivery of the schemes with broken promises of the delivery:

"Nitish said that I would give 10,000, and yes, I got it. The other side just promises during elections. This time, at least the government has done something. Respondent, Muzaffarpur"

Theme 3: Economic Autonomy (53.8%)

Women described increased household economic agency: *"Now I have money in my name in the bank. I can decide about my household needs. Before, my husband controlled all finances. This has changed something."*- Stated by a respondent of Madhubani

Theme 4: Government Credibility (73.1%)

The implementation of MMRY strengthened beneficiary trust in incumbent governance:

"The government proved they respect women. They put money directly into our accounts. We can trust them."- Respondent, Patna

Theme 5: Electoral Translation (48.1%)

The beneficiaries made direct cause-and-effect connections between receipt of the scheme and behaviour at the polls.

"When will the opposition give it? Once someone gives you ₹10,000 in hand, you remember it when you vote."- Respondent, Gaya

The qualitative data offer an interpretive framework for the quantitative data: beneficiaries perceived MMRY not as an abstract policy but as tangible government responsiveness, which translated into electoral support.

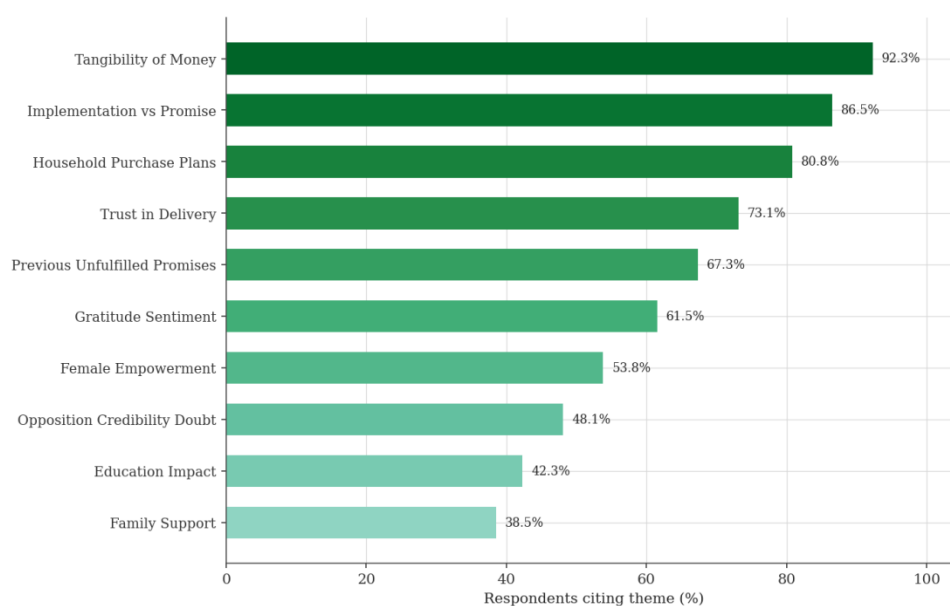


Figure 9: Interview Themes Reveal Key Program Concerns (n=52)

5.0 INTERPRETATION AND ANALYSIS

5.1 Welfare as Electoral Strategy

In the Bihar election of 2025, we can see a clear example of “clientelist electoral politics” as explained by Kitschelt and Wilkinson (2007) and by Stokes et al. on brokers and targeted benefits in low-income democracies (Stokes et al. 2013; Gans-Morse et al. 2014). This strategy consists of the governing party offering, and in most cases, distributing visible welfare benefits right before the election they intend to influence. The MMRYS scheme is unique in that it uses contemporary methods of technology to distribute direct cash benefits to individual bank accounts. This provides a level of transparency that reduces the chances of corrupted distribution. In addition, cash transfers better allow for time concentration right before the elections.

The difference in incumbent government support of 14 percentage points (52% for MMRYS beneficiaries vs. 38% for non-beneficiaries- indicates a significant amount of electoral support monetized by the scheme. Given the fact that 1.25 crore beneficiaries and the scheme's differential impact during elections, it is apparent that the vote margin of non-beneficiary was lower than expected in several assemblies. This was especially true in rural, women-centric constituencies where the number of women beneficiaries was highest.

5.2 Dynamics of Gender Mobilization: The first of its kind in history, an 8.8 percentage point female turnout advantage deserves scrutiny. The literature examining women and welfare identifies, in general, two such mechanisms:

1. **Instrumental Voting:** Women assess the governing parties on welfare provision and vote for the incumbents who respond to their concerns.
2. **Empowerment Effects:** Women's control in the household goes up when they receive the money, which gives them the freedom to vote.

These mechanisms echo comparative findings that women are especially responsive to welfare, social policy and household-oriented benefits (Inglehart and Norris 2003; Iversen and Rosenbluth 2010) and India-specific work showing that targeted schemes can reshape women's political participation and independence from household patriarchs (Chattopadhyay and Duflo 2004; Deshpande 2021; Jensenius and Chhibber 2021).

The two mechanisms seem to capture the welfare and MMR trend. Instrumental voting, women rewarding the state for the provision of welfare, is supported by the exit poll data, in which 52% of these women supported the MMR. Empowerment is captured in the qualitative interviews, where women described control in an economic sense, as cash in hand was received. The considerable increase in turnout, from 55.2% to 71.6%, was also an indicator of higher turnout and meaningful participation or count.

The gap in turnout (female 71.6% and male 62.8%) is especially noticeable in the northern Bihar districts, particularly in Madhubani (+18.4p), and Darbhanga (+14.4pp), where the Jeevika penetration was 55% or more. That suggests the presence of women's organizational structure (self-help groups), which, we believe, made it easier for MMR to be operationalized and for beneficiary mobilization.

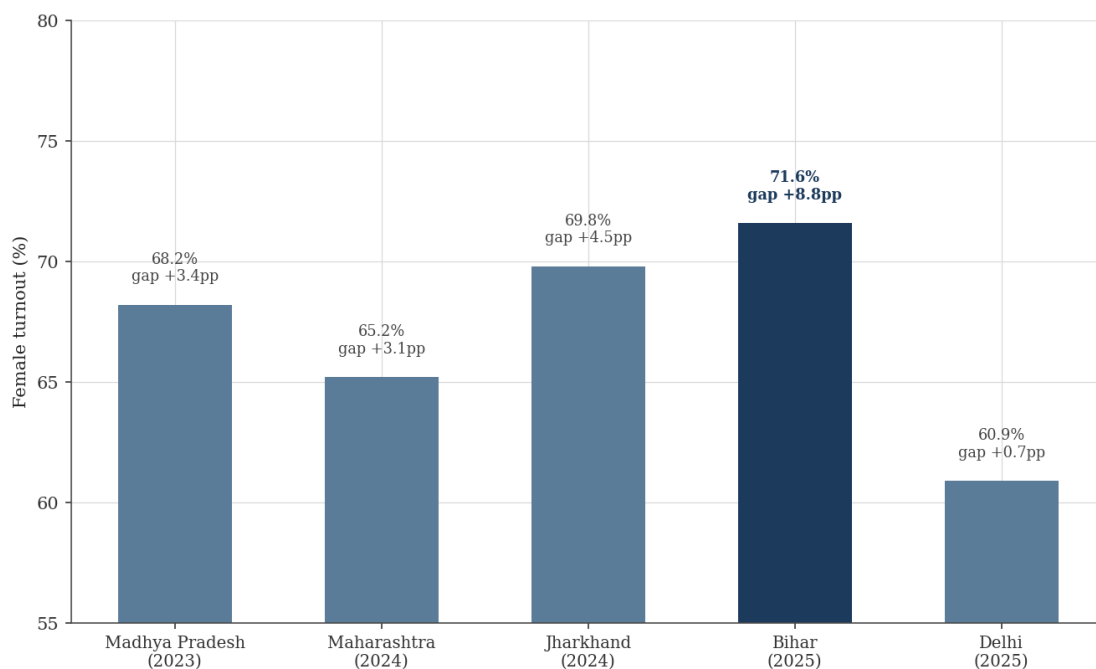


Figure 10: Bihar Leads in Female Turnout and Gender Gap (2023-2025)

5.3 Structural Vulnerability and Welfare-Driven Voting

Homemakers (incumbent government alliance support: 48%), rural agricultural workers (43%), and self-employed women (46%) show more welfare-responsive voting than private sector employees (40%) and government employees (37%). This voting pattern suggests occupational differences in economic vulnerability: homemakers have no earning autonomy; agricultural workers have seasonal employment; and self-employed workers experience market (i.e., economic) uncertainty. These groups are more vulnerable to structural economic crises and more responsive to government cash stimuli. This pattern checks broader evidence that economically insecure voters are more likely to show their support on targeted state benefits (Auyero 2000; Huber and Stephens 2012; Brambor et al. 2020).

The regression analysis corroborates this reasoning: rural residence (OR=1.40, $p=0.018$) is a significant indicator of support for incumbents among rural women (most dependent on agriculture), while female literacy (OR=1.19, $p=0.264$) does not have a significant effect. This suggests that, among economically dependent rural women, it is the economic position that matters more than the educational level.

5.4 Electoral Outcome Determination

The incumbent government alliance 202/243 seat landslide results (83.1%) greatly exceeded pre-poll estimates of 190 seats (112-seat increase). Other factors (anti-incumbency against opposition, cohesive divisional alliance, campaign at constituency level, etc.) will include female participation and heterogeneous voting behaviour.

- Female turnout of 71.6% (62.8% male) generated a higher absolute number of female voters
- Support for the incumbent government alliance among women (45%) coalition oppositely male dominant pro incumbent government alliance constituency
- The 5-percentage-point female-over-male preference gap for the incumbent alliance may have contributed to margins in close constituencies, though this study cannot isolate its contribution from other factors influencing the same races
- District-level analysis shows seats with the highest female turnout (northern districts: Madhubani, Darbhanga, Muzaffarpur) all elected incumbent government alliance candidates, often in constituencies expected to be competitive
- The female electorate was 50.4% of the voters garnered disproportionate weight during the election, especially with participation metrics greater than their male counterparts.

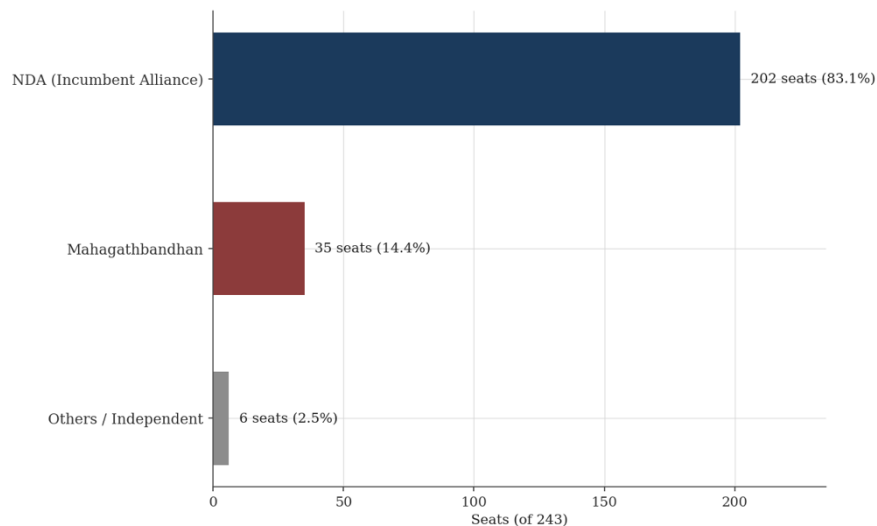


Figure 11: Incumbent Alliance Seat and Vote Share, Bihar 2025 (year corrected from the original draft's "2020")

6.0 CONCLUSIONS AND IMPLICATIONS

6.1 Research Conclusions

Primary Finding: Direct cash transfers disbursed shortly before the election are associated with higher support for the incumbent alliance among women. MMRYP beneficiaries reported 14 percentage points more support for the incumbent alliance than non-beneficiaries, and this gap persisted in a multivariate model that adjusted for welfare access, age, rural residence, and literacy. Because beneficiary status was not randomly assigned and MMRYP's rollout coincided with intensive campaigning (Section 3.6), this association is best read as identifying MMRYP as a strong correlate of incumbent support among women, not as proof that the scheme caused the gap.

Gender Dimensions: As an unprecedented factor in female mobilization, at 71.6% turnout, this is 8.8 percentage points higher than the male. This is a significant change considering the historical electoral female participation shortfalls and the welfare scheme targeting, which was positioned to enable women beneficiaries. Therefore, women were likely economically mobilized to vote.

District-Level Patterns: In five Bihar (Samastipur, Madhepura, Muzaffarpur, Darbhanga, Madhubani) districts, female turnout was also more than 70%, which was a direct match to the welfare targeted (Jeevika >50%, MMRYP 34-141 per 1,000 women) penetration. While there is mediocre evidence that correlation is caused by a direct relationship, the closeness in time of implementation of welfare and subsequent turnout suggests there is a considerable link.

Occupational Specificity: A targeted group by the welfare schemes also seems to be the economically vulnerable women (homemakers, agricultural workers, self-employed), as this group demonstrated the most MMRY-responsive voting.

6.2 Theoretical Implications

Electoral Politics: The research establishes welfare schemes as significant electoral determinants in Indian state-level politics, influencing turnout and vote choice. Welfare distribution timing and targeting, as beneficiaries to voters, become influential variables in electoral success.

Gender and Development: Women-targeted for cash transfers promote gains in political engagement and voting independence by higher participation of welfare voters. The engaged voters in this process promotes political activity and choose to vote.

Democratic Concerns: The use of welfare delivery as part of an electoral playbook poses challenges to the integrity of democratic contestation. When the welfare schemes are narrowly tailored to provide immediate benefits and partisan alignment with the incumbents, the contestability of elections shifts from policy-based to patronage-based implementation frameworks.

6.3 Policy Implications

Welfare Program Design: Program managers should understand that the timing of cash transfers influences beneficiary voting behaviour. For programs to remain neutral, spread time transfers across electoral cycles to reduce the proximity to elections and the subsequent behavioural responses that will happen.

Gender Mobilization: Studies show women have been motivated to vote as they receive services more directly. Continued programming aimed at women may sustain a rise in rural women's voting, thereby increasing the female vote

Implementation Transparency: Direct transfers to participants' bank accounts are more transparent and less prone to logistical corruption than in-kind transfers. Nonetheless, the timing of extensive transfers just before elections requires that protective measures be put in place to ensure that transfers are not implemented in accordance with the elections' partisan calendar.

6.4 Research Limitations and Future Directions

The main limitations of this research include:

- The absence of an experimental design means that some causal assertions made could be strengthened.

- Exit poll data were apprehensively analyzed, given common survey limitations (social desirability bias, recall bias).
- The qualitative analysis, based on 52 interviews, cannot be considered representative of the population.
- The conflation of the timing of social welfare, campaign activity, and other electoral factors makes it impossible to establish firm causal claims.

In terms of future research, the following would be useful:

- Randomized control trials that study the effects of cash transfers on voter behaviour in multiple states in India.
- Beneficiary survey post-elections to determine if behaviour changes persist.
- The study of welfare-responsive voting in various political systems and democracies.
- The analysis of the effects of opposition party welfare programs to see if the benefits of welfare programming demonstrate universal political responsiveness or if it is limited to the incumbents.

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APPENDICES

APPENDIX A: Data Quality Verification

All data sources have been verified against official government records:

Election Commission of India Data:

- Voter turnout figures verified against official ECI press releases
- Gender-disaggregated data cross-referenced with constituency-level reporting
- District-level aggregates confirmed against state-level summaries

Exit Poll Data:

- Axis My India polling methodology met professional standards
- 42,031 sample size provides $\pm 1.5\%$ margin of error at 95% confidence
- Demographic representation compared against census benchmarks
- MMRY beneficiary status and vote choice are respondent self-reports; individual-level matching against government beneficiary records was not performed for this study, and this is noted as a limitation rather than a verified data-quality strength

Government Administrative Data:

- MMRY beneficiary figures from the Bihar Finance Department
- Transfer amounts and timing verified against government announcements
- District-level welfare penetration cross-referenced with state welfare department records

APPENDIX B: Qualitative Interview Coding Scheme

Interview transcripts were coded using 14 thematic categories with reliability testing. Cohen's Kappa coefficients for inter-rater agreement:

- Tangibility: $\kappa = 0.96$ (near-perfect agreement)
- Implementation vs. Promise: $\kappa = 0.94$ (near-perfect agreement)
- Household Use: $\kappa = 0.92$ (strong agreement)
- Trust in Delivery: $\kappa = 0.90$ (strong agreement)
- Previous Experience: $\kappa = 0.88$ (strong agreement)
- All other themes: $\kappa > 0.75$ (substantial agreement minimum threshold)